

Selective discharge

What factors affects the species which is preferentially [O] or [R]?

$E^\ominus$ value	species with the most <i>favorable</i> value will be selectively discharged at cathode and anode
state	molten/ aqueous (consider H_2O if aqueous)
conc.	a more highly concentrated ion may be preferentially discharged even if its $E^\ominus$ value is less favorable
electrode	reactive electrodes may participate in electrolysis (graphite and platinum electrodes are inert)

How to deduce which species is preferentially [O] or [R]?

Step 1	Identify all possible species that can be oxidised/ reduced. • for [O] at anode: • for [R] at cathode:
Step 2	Compare $E^\ominus$ values from <i>Data Booklet</i> • for [O] at anode: select eq ⁿ with most _____ $E^\ominus$ • for [R] at cathode: select eq ⁿ with most _____ $E^\ominus$
Step 3	Consider concentration effects • concentration affects E values

Predict the products formed at the cathode and anode when $Zn(NO_3)_2$ (aq) is electrolysed using Zn electrodes. Describe what will be observed.

➤ List all species present:

Possible species that can be [O] at anode:

Possible species that can be [R] at cathode:

➤ Half-equations for reactions that occur:

[O]; anode:

[R]; cathode:

➤ Explain why Zn^{2+} is preferentially reduced at the cathode.

$E^\ominus(Zn^{2+}/Zn)$ is _____ than _____

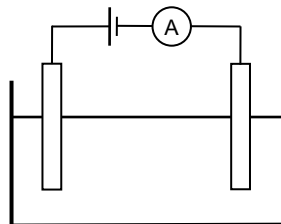
Hence, Zn^{2+} has a greater tendency to undergo reduction than H_2O .

➤ Observation

_____ decreases in mass

_____ is coated with Zn and increase in mass

Electrolytic cell setup



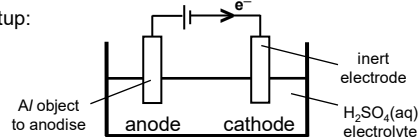
Recommended sequence to label electrolytic setup

- Label the positive and negative terminals of the battery
- Label the positive and negative electrodes
- Label the direction of the e^- flow
- Based on e^- flow, label anode (loses e^- for [O]) & based on e^- flow, label cathode (gains e^- for [R])

Industrial Applications

- Anodising of aluminium
(coating aluminium anode with Al_2O_3 layer)

➤ setup:



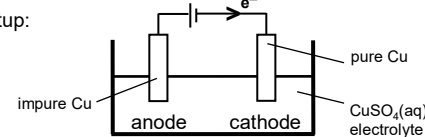
➤ key reactions:

anode: $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

$O_2(g)$ produced reacts with _____ to form _____

- Purification of copper through electrolysis

➤ setup:



➤ key reactions:

anode:

cathode:

Faraday's Law

$$Q = I \times t$$

Q = charge (C)

I = current (A)

t = time (s)

$$Q = n_e \times F$$

F = Faraday constant (96500 C mol^{-1})

n_e = no. of mol of e^- (mol)

Note: n_e refers to _____

$$Q = n_e \times L \times e$$

n_e = no. of mol of e^- (mol)

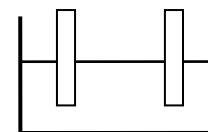
L = avogadro constant (6.02×10^{23})

e = charge per e^- ($1.60 \times 10^{-19} \text{ C}$)

Example

An electrolytic cell is setup using Zn electrodes immersed in $ZnCl_2(aq)$. What is the time required for 3.0 g of Zn(s) to be deposited onto the cathode given that the current is 3.0 A?

➤ Sketch labelled setup:



➤ no. of mol of Zn ($A_r = 65.4$) deposited:

➤ no. of mol of e^- required:

➤ charge required:

➤ time required:

Selective discharge

What factors affects the species which is preferentially [O] or [R]?

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state	molten/ aqueous (consider H_2O if aqueous)
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electrode	reactive electrodes may participate in electrolysis (graphite and platinum electrodes are inert)

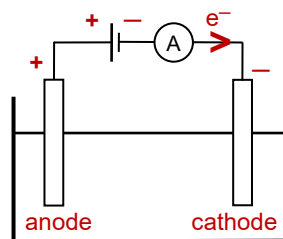
How to deduce which species is preferentially [O] or [R]?

Step 1	Identify all possible species that can be oxidised/ reduced. • for [O] at anode: anions, electrode, neutral species • for [R] at cathode: cations, neutral species
Step 2	Compare $E^\ominus$ values from <i>Data Booklet</i> • for [O] at anode: select eq ⁿ with most negative $E^\ominus$ • for [R] at cathode: select eq ⁿ with most positive $E^\ominus$
Step 3	Consider concentration effects • concentration affects E values

Predict the products formed at the cathode and anode when $Zn(NO_3)_2$ (aq) is electrolysed using Zn electrodes. Describe what will be observed.

- List all species present: **Zn electrodes, Zn^{2+} , NO_3^- , H_2O**
Possible species that can be [O] at anode: **Zn electrodes, NO_3^- , H_2O**
Possible species that can be [R] at cathode: **Zn^{2+} , H_2O**
- Half-equations for reactions that occur:
[O]; anode: **$Zn \rightarrow Zn^{2+} + 2e^-$**
[R]; cathode: **$Zn^{2+} + 2e^- \rightarrow Zn$**
- Explain why Zn^{2+} is preferentially reduced at the cathode.
 $E^\ominus(Zn^{2+}/Zn)$ is **less negative** than **$E^\ominus(H_2O/H_2)$**
Hence, Zn^{2+} has a greater tendency to undergo reduction than H_2O .
- Observation
anode decreases in mass
cathode is coated with Zn and increase in mass

Electrolytic cell setup



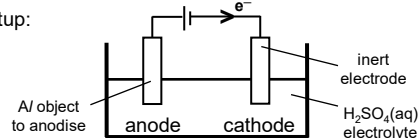
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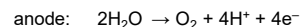
Industrial Applications

- Anodising of aluminium**
(coating aluminium anode with Al_2O_3 layer)

➤ setup:



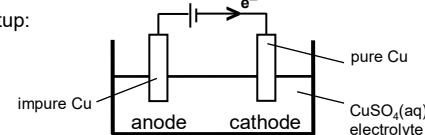
➤ key reactions:



$O_2(g)$ produced reacts with **Al** to form **Al_2O_3**

- Purification of copper through electrolysis**

➤ setup:



➤ key reactions:



Faraday's Law

$$Q = I \times t$$

Q = charge (C)
 I = current (A)
 t = time (s)

$$Q = n_e \times F$$

F = Faraday constant (96500 C mol^{-1})
 n_e = no. of mol of e^- (mol)

Note: n_e refers to **actual amount, not mol ratio**

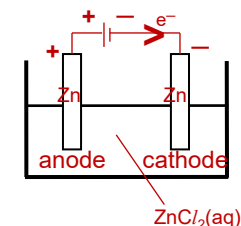
$$Q = n_e \times L \times e$$

n_e = no. of mol of e^- (mol)
 L = avogadro constant (6.02×10^{23})
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➤ Sketch labelled setup:



➤ no. of mol of Zn ($A_r = 65.4$) deposited:

$$3.0 / 65.4 = 0.0459 \text{ mol}$$

➤ no. of mol of e^- required:



$$n_e = 0.0459 \times 2$$

$$= 0.0917 \text{ mol}$$

➤ charge required:

$$Q = n_e \times F$$

$$Q = 0.0917 \times 96500$$

$$= 8850 \text{ C}$$

➤ time required:

$$Q = I \times t$$

$$t = 8850 / 3.0$$

$$= 2950 \text{ s}$$